

An Efficient Automated Framework for Wildlife Detection and Prevention in Agricultural Zones Nearer to Forest Reserves

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Abstract- Agriculture revenue is a significant contribution to the family of farmers. However, human-wildlife confrontation is a key challenge in agricultural regions located near the forest reserves. Mainly wild animals often enter the farmland and destroy the agricultural productivity, which results in severe financial loss, disruption of livelihoods, and even threats to farmer life. To minimize these issues, this article proposes a realistic animal intrusion detection and prevention model for agricultural regions. The proposed scheme leverages superior methods to address these innovative solutions to exploit to identify the movement of wild animals via live video feed and send cautioning message alerts to prevent encounters, and thus the protection of both crops and farmers. The system implements YOLO v5, CNN, and YOLO v8. The IoT hardware components include Arduino Uno, ESP8266, NodeMCU, LCD display, and buzzer. It notices wild animals in real time, examines their movement patterns, and triggers automated deterrent mechanisms. The system ensures proactive wildlife management while minimizing harm to both animals and farmers. Additionally, alert systems that notify farmers and local authorities via SMS to enable timely response and prevent crop damage. This new solution efficiently bridges the gap between technology and conservation, promoting a harmonious coexistence between humans and animals while ensuring sustainable agricultural practices.

Keywords: wild animal detection, IoT, real-time monitoring, conservation, machine learning, crop protection.

1. Introduction

The growing spread of human settlements and development has inevitably increased human-wildlife conflicts (HECs) across the farmland near the forest regions. Agriculture is a key occupation for the income of many people. Deforestation in several parts of the forest regions has led to people establishing livelihoods and cultivating the land. As a result, when animals face shortages of water, food, and shelter in the forest, they venture into nearby farmland in search of resources. This leads to HECs, causing crop damage and even human casualties. Occasionally, farmers resort to killing wild animals, sometimes making electric

fences or poisoned food. These conflicts create financial losses for farmers, who lose their investments in crops as shown in figure 1. Farm owners with larger-scale land face even greater challenges in protecting their extensive fields from animal intrusions, making farming increasingly difficult. To mitigate and evade HEC, some advanced and traditional schemes have been employed. The traditional systems typically include guarding crops, chasing animals away with noise like drum-beating, firecrackers, and shouting, and setup physical barriers such as electric fences, use acoustic and light-based deterrents. The various advanced technological developments to address this issue include early detection and warning systems to track animal movements, as well as the internet of thing (IoT) combined with surveillance techniques. However, existing wildlife monitoring methods, including human surveillance and motion-triggered cameras, face several limitations, including high operational costs, slow response times, and limited scalability. Recent advances in AI and the IoT are creating new opportunities for real-time wildlife detection and monitoring, improving conservation approaches while reducing reliance on human involvement.

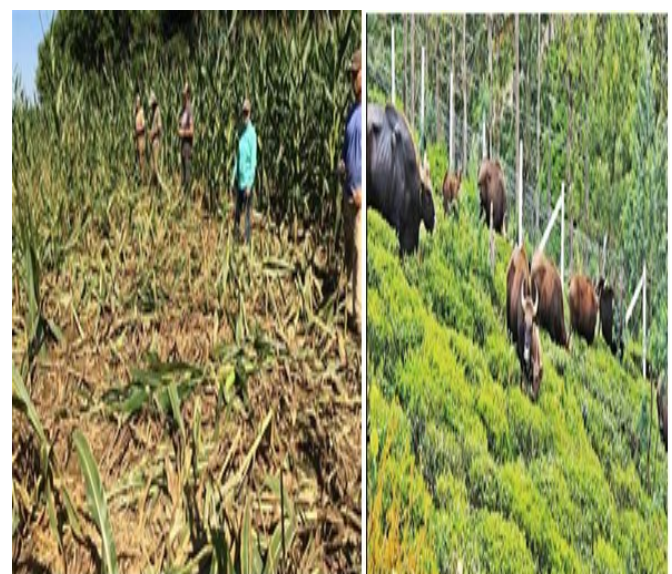


Figure 1 Animal intrusion in farm lands and crop damages.

The combination of AI-enabled object recognition, IoT-empowered smart sensors, and data analytics enables efficient automated detection of animals in their natural habitat. Machine learning prediction models, such as YOLO v8, YOLOv5 and convolutional neural networks (CNNs), improve the accuracy of species identification precision and movement tracking capabilities. Edge computing further enhances system performance by processing data locally before transmitting only pertinent information to cloud servers, thereby reducing latency and power consumption. This paper introduces a comprehensive IoT- and YOLO v8 based wildlife detection system that enables real-time monitoring, notifies conservationists about potential threats such as poaching, and facilitates long-term data collection for ecological research.

2. RELATED WORKS

Animal intrusion leads to major devastation in agricultural crops and lands across the forest regions. Several mechanisms have previously been proposed to prevent the intrusion of wild animals. The combination of IoT setups with machine learning plays a major task in solving this issue. In this part, we review the conventional animal detection models and technologies that have been addressed for the detection and deterrence of endangered and harmful animals. A deep learning-based energy-efficient automatic wild animal detection scheme was proposed [1]. Energy reduction is carried out by edge devices. A cross-scale feature aggregation network-aided animal identification is described [2], a time-frequency map is employed for animal identification, and CNN is used to classify humans from animals. In [3,4], proposed the YOLOv8 to track and monitor wild animals. This model detects four types of wild animals, for examples tigers, leopards, lions, and bears. In [5], camera traps are utilized to automatically identify wildlife. The performance of the animal detection models is validated by day-night joint training. Unmanned aerial vehicle-based object detection is proposed to detect wild animals with integrated deep features and shallow features [6,7]. Ultrasonic sensors are employed to assist in the wild animals intrusions detection on agricultural fields [8]. The farmer gets an alert message via the IoT platform. The authors [9,10] presented an inventive animal intrusion detection model implementation of Yolov8. It focuses on the issues of wild animal intrusions in agricultural farms with precise animal detection and the activation of suitable warning messages. Crop production is severely destroyed by animal attacks, which have consequences for the farmers revenue and food safety [11].

This model details how this problem arises via the combination of IoT and deep learning techniques. Deep learning methods are supported for animal detection and categorization. In [12], a novel result that harnesses the power of edge deep learning techniques via IoT for wildlife intrusion monitoring and deterrence schemes is addressed. The presented model is implemented to make remote monitoring to safeguard vast crop fields. For animal intrusion detection and categorization via an AI-CAM with deep learning and laser detection methods are suggested. This model also guarantees effective animal deterrence and live monitoring for farmers, allowing them to evaluate conditions with the help of a smart IoT-based rover. In [13],

the YOLOv8-based method for wildlife animal detection was addressed. It used super-pixel-based fast fuzzy C-means to enhance wildlife animal monitoring abilities. Farmers create tough choices, like setting up electric fences to safeguard their crops, even though such methods can endanger animals crucial for ecological balance. To address these pressing challenges, [14,15] proposes a novel approach using an object-detection system.

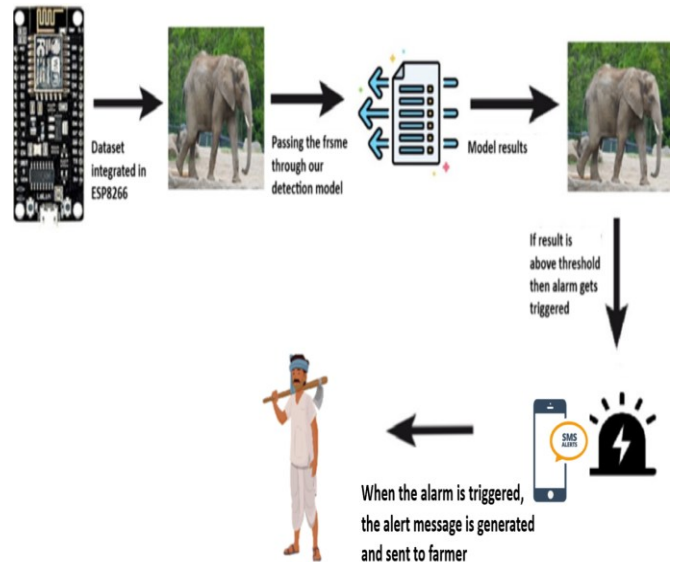


Figure 2 Smart animal intrusion detection and alert framework.

Specifically, a system was deployed based on the ESP32-CAM unit integrated with the YOLOv8 object-recognition model[16]. This model aims to predict endangered species and dangerous animals in agricultural settings, offering practical notifications to farmers and at-risk wildlife through a cloud-based intimation mechanism. In [17], it addresses the issue of harms to agricultural output and resulting fatalities among both humans and animals, including elephants. The residents and forest officials can be alerted promptly to avoid HWCs. A technique in image and audio processing employed an animal intrusion detection scheme addressed occurrence of wild animals near human living areas [18,19]. A computer vision system automatically captures wild animals[20,21]. This scheme used YOLO model to make predictions regarding the presence of wild animals. However, in remote areas sending alert notifications to officials depends on internet or mobile networks, which can be inoperable[22].

3. PROPOSED METHODOLOGY

The proposed AI- and IoT-based wildlife detection system embeds smart sensors, deep learning models, and endpoints processing that enhance real-time monitoring, improve detection accuracy, and lessen false positives. The proposed system model depicts as figure 2. The proposed model follows a multi-layered approach, comprising information acquisition, edge processing, cloud integration, and alert operations. In the data acquisition layer, various sensors such as passive infrared (PIR) motion sensors, thermal imaging cameras, and ultrasonic sensors are deployed in wildlife areas to detect movement, heat signatures, and animal sounds. These sensors help capture multimodal data, ensuring accurate detection in different

environmental conditions. The collected data is then processed at the edge to streamline response and enhance efficiency. This system incorporates AI, the IoT, and computer vision technologies to detect, monitor, and obstruct wild animals efficiently as depicted in figure 3. The methodology explores a structured approach to ensure accurate wildlife monitoring, real-time detection, and automated inhibition strategies, making it an effective tool for agricultural protection and sustainability actions.

The flowchart illustrates the complete workflow of the AI-powered wildlife detection and deterrence system. It highlights all eight key stages involved in system development, from initial data collection to final deployment and continuous improvement. An additional step has been included in the software deployment and testing phase to showcase tools used for performance validation and simulation. The system begins with data preparation, including processes like data sourcing, labeling, and augmentation, which are essential for training the model for real-time wildlife detection. Once the detection model is ready, the workflow branches into three core areas: hardware integration, movement monitoring, and alert implementation. In the hardware integration section, components like the Arduino uno, ESP8266 NodeMCU, an ultrasonic buzzer, and other hardware modules are assembled to support the detection system. The movement monitoring section involves motion tracking, behavioural analysis, and data visualization to understand animal. Activity of animal detection alerts more precisely are sent to farmers, enabling them to prevent crop damage caused by wild animals. The detailed overview of proposed a real-time animal intrusion detection and prevention system is shown in figure 4.

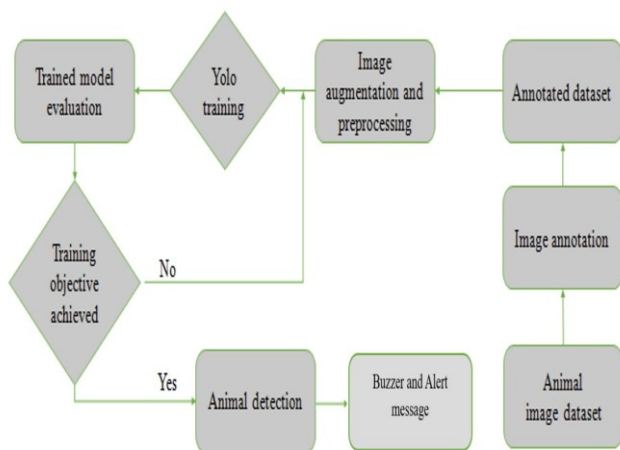


Figure 3 Functional blocks of a real-time animal intrusion detection and prevention system.

Meanwhile, the alert system is responsible for sending notifications, integrating the alert feature into a mobile app, and conducting field testing followed by system evaluation. The figure 5 is shown as the proposed classification model for a practical animal intrusion detection and prevention model. All three paths eventually converge into the final phase, which includes software deployment, further field testing, and system integration. The model undergoes AI refinements and firmware updates for continuous improvement, ensuring enhanced performance and reliability. Each step is essential for

building a robust, real-time animal detection and alert system.

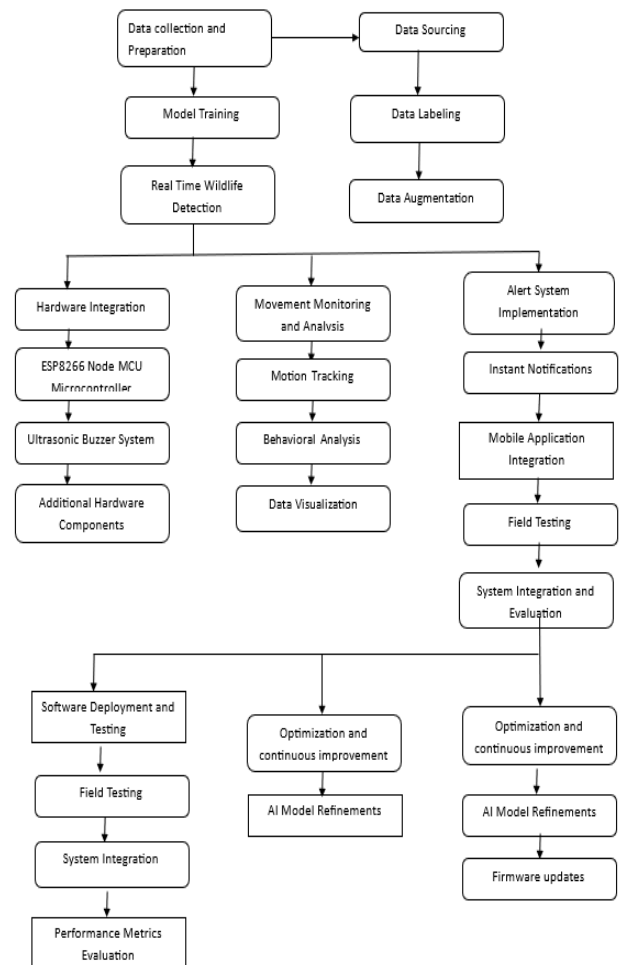


Figure 4 Real-time animal intrusion detection and prevention model

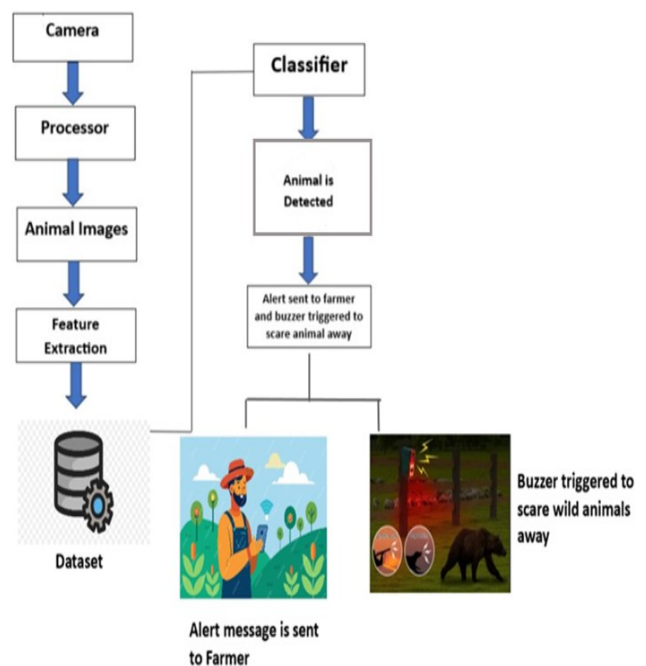


Figure 5 Typical functioning model of an animal intrusion detection and prevention scheme.

4. PERFORMANCE EVALUATION AND DISCUSSION

This model performs a real-time animal detection and generates alarm for the protection of form from the wild animals and save the human. The proposed model can detect wild animals, such as deer, elephant, tiger, leopard and bear. To validate the data reliability, Kaggle datasets is widely used. The proposed model used as totally 1850 images, based on this dataset into train, test, and validation sets of 20%, 20%, and 80% split respectively. 150 epochs were performed. To avoid false prediction, a confidence threshold is set to 0.69. It uses 0.01 learning rate. The proposed system model validates the following classifiers such as YOLOv8, CNN and YOLOv5. YOLOv8 performs better accuracy as compared to other existing models. The performance of the proposed models is determined animal intrusion detection using precision, Recall and F1-Score. The confusion matrix aids interpretation by highlighting the specific sentiment misclassifications made by each model. The simulation results showcase the performance of the proposed YOLOv8 model in comparison to the CNN and YOLOv5 models. Table 1 presents the performance metrics of the proposed model.

Models	Accuracy (%)	Precision (%)	Recall (%)	F-Score (%)	Execution time (ms)
YOLO v 8	90	90	94	92	150
CNN	86.5	80	79	79	210
YOLOv5	84.8	79	76	78	172

Table 1. Performance Metrics

The confusion matrix is typically used to assess the performance of a proposed classification scheme. The estimation validates the performance of each classes, which shows diverse performance metrics of the model such as accuracy, recall, precision and F1 score. These measures help assess the model's ability to correctly and incorrectly identify outcomes, providing a clear evaluation of its reliability. Table 2 presents the confusion matrix generated for the proposed system. These measures help assess the model's ability to correctly and incorrectly identify outcomes, providing a clear evaluation of its reliability. Table 2 presents the confusion matrix generated for the proposed system.

		Prediction class	
Actual class	Animal	1672	178
	Non animal	165	1453

Table 2. Confusion matrix

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{1672}{1672 + 178} = 90$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{1672}{1672 + 165} = 91$$

$$(TP) = 1672; (FP) = 178; (FN) = 165$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{(90 \times 91)}{(90 + 91)} = 90.49$$

Accuracy Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN} = \frac{1672 + 1453}{1672 + 165 + 178 + 1453} = 90$$

$$(TP) = 1672; (TN) = 1453; (FP) = 178; (FN) = 165$$

where

TP = True Positives (correctly detected animals)

FP = False Positives (false alarms)

FN = False Negatives (missed detections)

TN = True Negatives (correctly identified no-animal scenarios)

The performance of the proposed AI- and IoT-enabled wildlife detection system was gauged in diverse environmental settings, including dense forests and open landscapes. Figures 6 and 7 evaluate the proposed system performance. The system demonstrated an accuracy of over 92% in animal detection, significantly outperforming traditional camera traps, which averaged around 75%. The use of IoT-based sensors, such as PIR motion detectors, thermal cameras, and acoustic sensors, enhanced detection efficiency, particularly in low-visibility conditions such as nighttime monitoring. Figure 8 illustrates the balance between false negatives and false positives that the F1 score replicates. The proposed YOLOv8 performance is supported by F1 score measures of precision and recall, particularly in imbalanced datasets. At a 0.5 confidence level, YOLOv8 achieves a higher F1 score of 0.9. This indicates that the suggested scheme works much better than the traditional method. Because any missed identification can lead to serious crop damage, the proposed wildlife detection technique is a more effective solution. Figure 9 shows precision-recall graph, which presents typical in wildlife prediction tasks.

The YOLOv8 scheme conducts precision and recall in different datasets, where few animals were unpredicted. This plot indicates the balance between precision and recall, it reveals superior to properly detect wildlife exclusive of noteworthy overlap or misclassification. This model demonstrates that the algorithms prediction is greatly localized and accurate for detecting wild animals. The precision-confidence graph is shown in Figure 10. As the confidence threshold rises, precision increases. A precision of one at a threshold of 0.8 for all classes, demonstrates that even when greater confidence is needed, the YOLOv8 consistently validates precise estimations. This is especially vital for wildlife detection because false positives cause unnecessary deterrent measures that can upset regional ecosystems. The YOLOv8 model lowers the possibility of false positives while guaranteeing dependable system functionality improvement by ensuring that repellent activation only happens when the model is extremely confident in its prediction by maintaining a precision of 1.00 at higher thresholds.

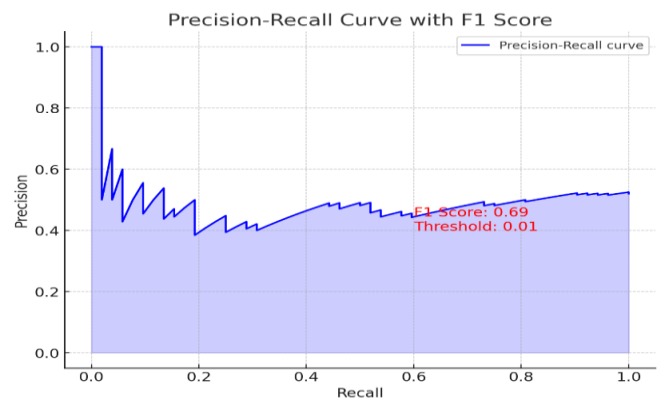


Figure 6 Precision-Recall curve with F1 score.

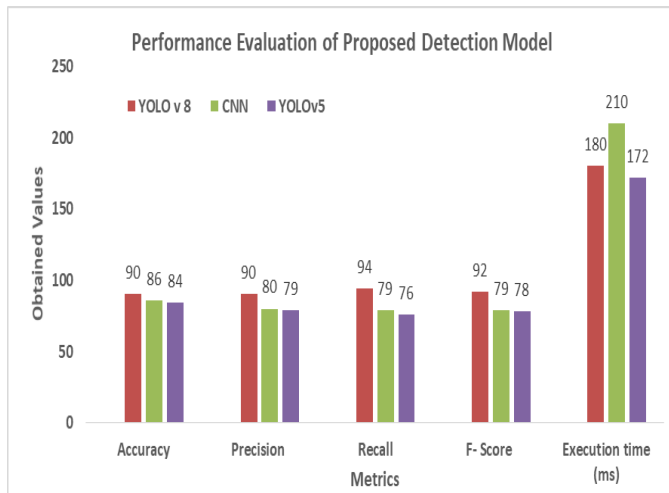


Figure 7 Performance comparisons of YOLOv8, YOLOv5 and CNN.

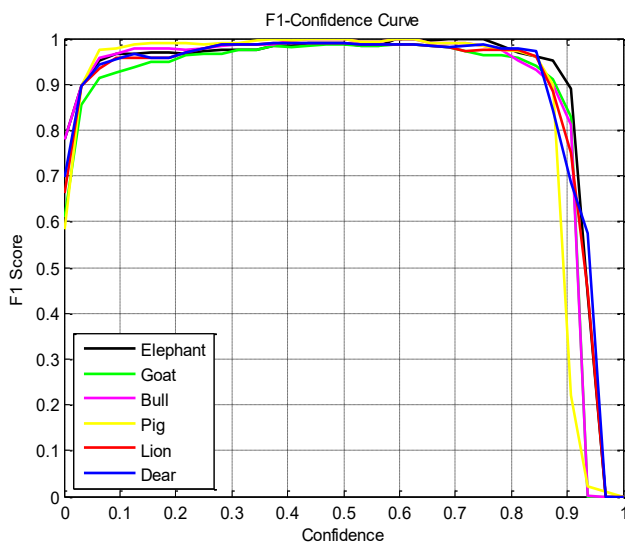


Figure 8 F1- Confidence plot of YOLO v8

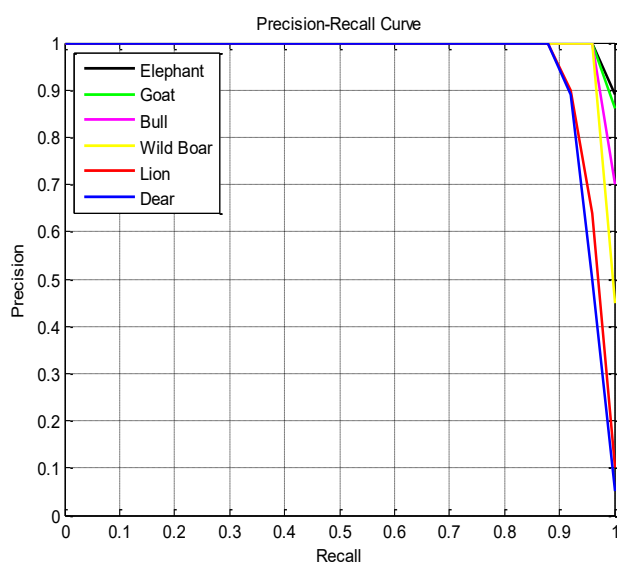


Figure 9 Precision- Recall plot of YOLO v8

By leveraging edge computing, data processing speed improved by 40%, reducing latency issues commonly faced

in cloud-dependent models. The system effectively differentiated between wildlife and human intruders, achieving an 90% accuracy in identifying poaching threats. Real-time notifications were successfully transmitted to conservation teams, enabling faster interventions and enhancing protection measures.

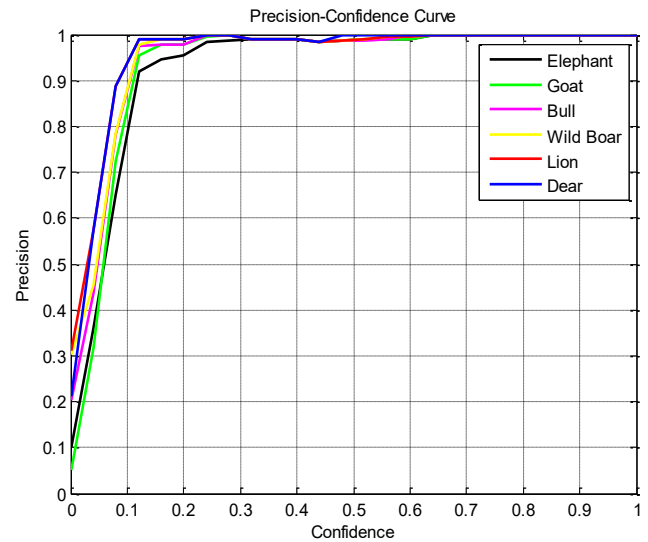


Figure 10 Precision-Confidence plot of YOLO v8

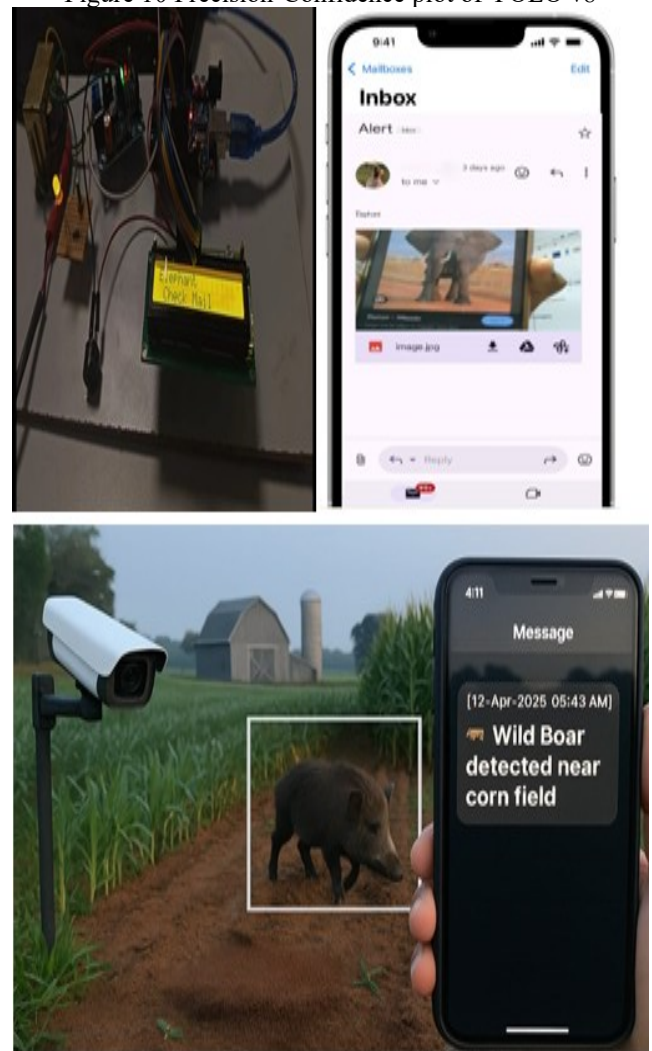


Figure 10 Hardware model with alert message sent to the farmer.

6. CONCLUSION

Human-animal conflict is a key issue in agricultural zones, especially in hill regions. This conflict has an impact on yield loss, crop field damage, loss of livestock, farmer injury, and even farmer loss of life. To overcome this issue, this article proposed an automated wildlife detection and prevention system for agricultural wildlife conflict areas. This model implements a machine learning algorithm with an IoT platform that detects and prevents a wild animal from entering the agricultural fields. The proposed system model demonstrates a significant potential mitigation of human-wildlife conflicts. This system model is used to enhance the accuracy by YOLOv8 and OpenCV, which provide effective real-time detection. This model also integrates with deep learning models such as transformers and vision-based neural networks that can further improve detection accuracy, especially in challenging environments like dense vegetation and low-light conditions. This autonomous deterrent system can be improved by incorporating AI-driven mechanisms. For instance, using reinforcement learning, the system analyzes past intrusion patterns and optimizes deterrent strategies (sound alarms, flashing lights, or automatic drones) based on the specific animal detected. Finally, integrating a mobile application with real-time alerts, AI-powered recommendations, and predictive analytics would empower farmers with actionable insights, helping them safeguard their crops more effectively.

References

1. Reon Sato; Hiroshi Saito; Yoichi Tomioka; Yukihide Kohira, "Energy Reduction Methods for Wild Animal Detection Devices", IEEE Access, 28 February 2022, 2169-3536.
2. Min Bao; Fu Zou; Mengdao Xing; BoyangJia, "A Cross-Scale Feature Aggregation Network Based on Channel-Spatial Attention for Human and Animal Identification of Life Detection Radar", IEEE Geoscience and Remote Sensing Letters, 24 February 2023,
3. R. Gayathri, P. H. Princy, and P. Krishandhini, "Wild animal intrusion detection system using YOLOv8," Int. Res. J. Modernization Eng. Technol. Sci., vol. 5, no. 5, pp. 4583-4586, May 2023.
4. M. Tan, W. Chao, J.-K. Cheng, M. Zhou, Y. Ma, X. Jiang, J. Ge, L. Yu, and L. Feng, "Animal detection and classification from camera trap images using different mainstream object detection architectures," Animals, vol. 12, no. 15, p. 1976, Aug. 2022.
5. G. Wang, Y. Chen, P. An, H. Hong, J. Hu, and T. Huang, "UAVYOLOv8: A small-object-detection model based on improved YOLOv8 for UAV aerial photography scenarios," Sensors, vol. 23, no. 16, p. 7190, Aug. 2023.
6. P. K. Panda, C. S. Kumar, B. S. Vivek, M. Balachandra and S. K. Dargar, "Implementation of a Wild Animal Intrusion Detection Model Based on Internet of Things," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), Coimbatore, India, 2022, pp. 1256-1261
7. K. M, B. V and A. K, "Animal Intrusion Detection Using Yolo V8," 2024 10th International conference on advanced computing and communication systems (ICACCS), Coimbatore, India, 2024, pp. 206-211,
8. A. K. Thomas, P. Poovizhi, M. Saravanan and K. Tharageswari, "Animal Intrusion Detection using Deep Learning for Agricultural Fields," 2023 5th International conference on smart systems and inventive technology (ICSSIT), Tirunelveli, India, 2023, pp. 1021-1027.
9. Anandaraj Mahalingam, Ganeshkumar Perumal, Gopalakrishnan Subburayalu, Mubarak Albathan, Abdullah Altameem, Riyadh Saleh Almakki, Ayyaz Hussain and Qaisar Abbas, "ROAST-IoT: A Novel Range-Optimized Attention Convolutional Scattered Technique for Intrusion Detection in IoT Networks" sensors 23, no. 19: 8044.
10. Lakshmi Palaniappan and K. Selvaraj, 'Profile and Rating Similarity Analysis for Recommendation Systems Using Deep Learning', Computer Systems Science & Engineering, (November 2021), Tech Science Press, DOI:10.32604/csse.2022.020670, vol. 41, no. 3, pp. 903-917.
11. Anandaraj M, Ganeshkumar P, Naganandhini S, Selvaraj K. A novel fuzzy programming approach for piece selection problem in P2P content distribution network. PeerJ Comput Sci. 2024 Jan 3;10:e1645.
12. K. V. Reddy et al., "Edge AI in Sustainable Farming: Deep Learning-Driven IoT Framework to Safeguard Crops From Wildlife Threats," in IEEE Access, vol. 12, pp. 77707-77723, 2024.
13. J. Chappidi and D. M. Sundaram, "Novel Animal Detection System: Cascaded YOLOv8 With Adaptive Preprocessing and Feature Extraction," in IEEE Access, vol. 12, pp. 110575-110587, 2024,
14. Raiaan, M.A.K., Fahad, N.M., Chowdhury, S., Sutradhar, D., Mihad, S.S. and Islam, M.M., 2023. IoT-based object-detection system to safeguard endangered animals and bolster agricultural farm security. Future Internet, 15(12), p.372.
15. Bhagabati, B., Sarma, K.K. and Bora, K.C., 2024. An automated approach for human-animal conflict minimization in Assam and protection of wildlife around the Kaziranga National Park using YOLO and SENet Attention Framework. Ecological Informatics, 79, p.102398.
16. S Naganandhini, D Shanthi. "Temporal fusion transformer-based strategy for efficient multi-cloud content replication" PeerJ Comput Sci. 2025 Mar 25;11:e2713. doi: 10.7717/peerj-cs.2713.
17. A. P E, A. C K, M. Shibili, R. C K and N. K, "Development and Implementation of an Animal Intrusion Detection System Using Image and Audio Processing," 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-7
18. M. -H. Ngo, H. -H. Do, H. -S. Nguyen, H. -D. Pham and Q. -H. Do, "Anomaly Detection based on Frequent Pattern Mining in Smart Home Devices," in IEEE Access, doi: 10.1109/ACCESS.2025.3603558.
19. P. Panindre, S. Acharya, N. Kalidindi and S. Kumar, "Artificial Intelligence-Integrated Autonomous IoT Alert System for Real-Time Remote Fire and Smoke Detection in Live Video Streams," in IEEE Internet of Things Journal.
20. N. Umapathi, S. Gajjala and S. Vennampalli, "An IoT-based Intelligent System for Real-time Monitoring of Animal Health and Geolocation for Enhanced Surveillance and Farm Management," 2025 International Conference on Sensors and Related Networks (SENNET) Special Focus on Digital Healthcare(64220), Vellore, India, 2025, pp. 1-6, doi: 10.1109/SENNET64220.2025.11136070.
21. Lakshmi Palaniappan and K. Selvaraj, "Profile and Rating Similarity Analysis for Recommendation Systems Using Deep Learning", Computer Systems Science & Engineering, Tech Science Press 2022, 41(3), 903-917, vol. 41, no. 3, pp. 903-917.
22. Anandaraj, M., Selvaraj, K., Ganeshkumar, P., Rajkumar, K., and Sriram, K., "Genetic Algorithm Based Resource Minimization in Network Code Based Peer-to-Peer Network", Journal of Circuits, Systems and Computers, Vol.30 Issue 6 2020(SCI Indexed).